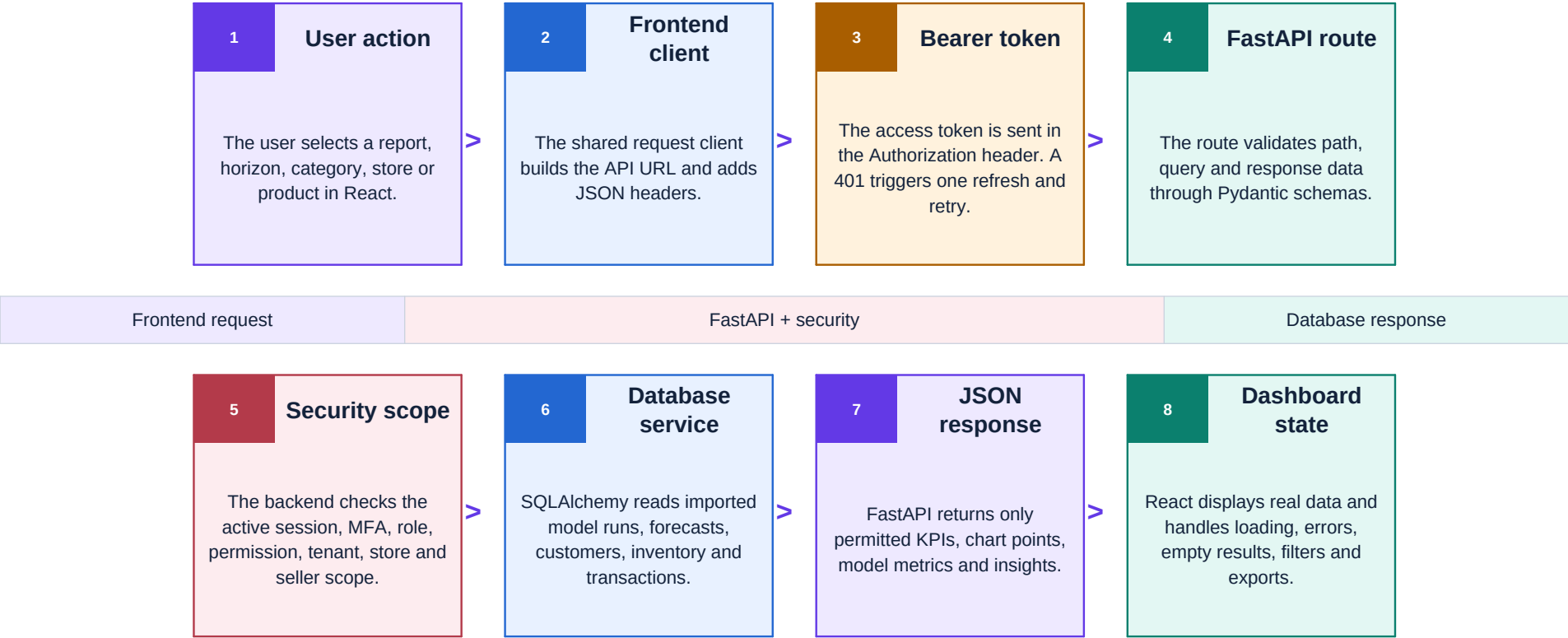


MarketMind AI - API Integration in Milestone 2

How the React frontend, secured FastAPI backend, database and machine-learning results work together.

1. End-to-end request workflow



2. What the integration guarantees

No direct database access The browser communicates only with FastAPI.	Backend-enforced RBAC Hidden UI controls are not treated as security.	Scoped data Business, store and seller records are filtered server-side.	Stable contracts Pydantic response schemas keep frontend fields predictable.
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Milestone 2 API Map and Swagger Demo

The main endpoints that connect model outputs to each authorised dashboard.

3. Main integrated endpoints

Feature	Endpoint	Role and scope	Dashboard result
Login	POST /api/v1/auth/login	All roles; Admin also submits MFA	Access + refresh token
Dashboard access	GET /api/v1/dashboard/access	Authenticated user	Allowed modules/actions
Segmentation summary	GET /api/v1/customer-segments/summary	Owner/Admin business; Manager store summary; Executive assigned	Segments, engagement and behaviour KPIs
Segment customers	GET /api/v1/customer-segments	Owner/Admin business; Executive assigned customers	Search, filter and customer rows
Revenue forecast	GET /api/v1/forecasts/revenue	Business Owner or Administrator	7/14/30-day INR forecast and metrics
Personal forecast	GET /api/v1/forecasts/personal	Signed-in Executive; Admin-selected seller	Seller-only forecast series
Demand forecast	GET /api/v1/forecasts/demand	Assigned Store Manager; Admin-selected store	Products, predicted demand and stock risk
Model monitoring	GET /api/v1/forecasts/monitoring	Administrator with MFA	Model versions, health and jobs

4. Swagger UI demonstration

1. Open 127.0.0.1:8001/api/v1/docs	2. Login Run POST /auth/login with email, password and Admin MFA when required.	3. Authorize Copy access_token, press Authorize, and paste the token only.	4. Test Run one role-allowed forecast. Swagger shows request, response and status.
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Short speaking content

For API integration, our React frontend uses one shared API client. Whenever the user opens a report or changes a filter, the client sends a request to FastAPI with the JWT access token. FastAPI validates the input and then checks the active session, role, permission and business, store or seller scope. Only after these checks does SQLAlchemy read the imported model results from the database. The backend returns a structured JSON response containing KPIs, forecast points, model metrics and insights, which React displays in the dashboard. If the access token expires, the client uses the refresh token once and retries the request. Swagger is generated automatically from the FastAPI routes and Pydantic schemas, so we can authorize, test the real endpoints and view their request and response contracts without using the frontend.

Useful status meanings: **200** success, **401** missing/expired authentication, **403** authenticated but not permitted, **404** no matching imported result, and **422** invalid query or request data.